Quantum Computing & AI

2025-2026 Technology Report

Breakthroughs, Trends & Global Economic Impact

1. Quantum Computing Breakthroughs

1.1 Google 13000x Speedup

October 2025: Google Quantum AI achieved 13,000x speedup over classical supercomputers in physics simulation. Published in Nature, this milestone demonstrates quantum supremacy transitioning from theory to practice.

1.2 China Three-Phase Strategy

Pan Jianwei outlined China's quantum roadmap: Phase 1 (complete) proof of concept; Phase 2 (2025) specialized quantum simulator for quantum chemistry and superconductivity; Phase 3 (goal) universal quantum computer.

1.3 Quantinuum Helios

November 2025: Quantinuum launched Helios quantum computer featuring ion-trap technology with high-fidelity gates, integrated quantum ML library, and hybrid quantum-classical workflow support.

2. AI Frontiers

2.1 AI-Quantum Convergence

By March 2026, China's daily AI token usage exceeded 140 trillion, 1000x growth from early 2024. Classical computing limits are being reached. Google and Oratomic research suggests AI-assisted quantum computing may break encryption sooner than expected.

2.2 Explainable AI

NSFC launched the 'Explainable Next-Gen AI' program, combining neural network quantum states, tensor networks, and generative models into a novel computational framework.

2.3 Brain-Computer Interface

2025: Brain-computer interface enabled patients to speak and sing with emotion for the first time, marking a new height in neuroscience-AI convergence.

3. Investment & Commercialization

McKinsey Quantum Technology Monitor 2025 reports record investment levels:

Direction	Share	Description
Quantum Hardware	~45%	Qubits, controllers, cooling
Quantum Comms	~25%	QKD, quantum cryptography
Quantum Sensing	~15%	Precision measurement, navigation
Quantum Software	~15%	Algorithms, simulation platforms

4. Global Economy 2026

WTO cut 2026 global trade growth forecast to 0.5% (from 2.4% in 2025). IMF/OECD predict global GDP growth of 2.7%-3.1%. The world faces four major shifts: geopolitical conflicts, trade protectionism, tech polarization, and social inequality.

NVIDIA market cap exceeded $5 trillion in October 2025, surpassing Microsoft and Apple. China's R&D spending intensity reached 2.8%, exceeding OECD average for the first time.

Key trends: AI-driven productivity gains, quantum computing commercialization, renewable energy transition, cybersecurity threats, and supply chain restructuring.